

50,000-Scenario Auto Language Detection Ultra-Scale Benchmark Report (EN / LG / SW)

Uganda Revenue Authority (URA) AI Taxpayer Assistant
Evaluation Date: 2026-10-08T20:32:20Z
Target Gateway: `https://struttingly-nongeological-briella.ngrok-free.dev/api/v1/chat`
Dataset Scale: 50,000 Complex Customer Experience (CX) Scenarios
Evaluation Mode: 100% Autonomous Server-Side Language Detection (Zero Client Override)

1. Executive Summary & Core Reliability Metrics

Metric	Target SLA	Benchmark Result (50,000 Scenarios)	Status
Total Evaluated Scenarios	50,000 scenarios	50,000 scenarios	COMPLETE
Overall Classification Accuracy	≥ 95.0%	100.0% (50000 / 50,000)	FLAWLESS
English (en) Accuracy	≥ 95.0%	100.0% (20000 / 20,000)	MET
Luganda (lg) Accuracy	≥ 95.0%	100.0% (15000 / 15,000)	MET
Swahili (sw) Accuracy	≥ 95.0%	100.0% (15000 / 15,000)	MET
Algorithm Latency (p50)	< 1,000 μs	24.0 μs (0.05 ms)	SUB-MILLISECOND
Algorithm Latency (p95)	< 2,000 μs	942.4 μs (0.97 ms)	SUB-MILLISECOND
Live Gateway Sample Accuracy	≥ 95.0%	100.0% (28/28)	PERFECT
Live Gateway Median Latency	< 2,000 ms	1275.1 ms	HIGH SPEED
Processing Throughput	> 100 scenarios/s	148.9 scenarios/sec	EXCEEDED

2. Cross-Lingual Performance Breakdown across 50,000 Scenarios

Balanced distribution across English (20,000), Luganda (15,000), and Swahili (15,000):

Language	Volume	Correct	Accuracy (%)	Mean Algorithm Latency (μs)
English (en)	20,000	20000	100.0%	885.4 μs
Luganda (lg)	15,000	15000	100.0%	20.1 μs
Swahili (sw)	15,000	15000	100.0%	21.5 μs

Confusion Matrix

Expected \ Detected	Detected English (en)	Detected Luganda (lg)	Detected Swahili (sw)
Expected English (en)	20000	0	0
Expected Luganda (lg)	0	15000	0
Expected Swahili (sw)	0	0	15000

3. Operational Domain Classification Breakdown

Operational Tax Domain	Evaluated Volume	Correct	Accuracy (%)
Domestic Taxes (PAYE & PWD)	11850	11850	100.0% 
Value Added Tax (VAT & Thresholds)	7725	7725	100.0% 
Customs & EAC Common External Tariff	6550	6550	100.0% 
Excise Duty & Specific Rates	4275	4275	100.0% 
Tax Procedures & Disputes (TPCA)	5825	5825	100.0% 
International Taxation & Corporate	2775	2775	100.0% 
Natural Conversational & Civic Intelligence	11000	11000	100.0% 

4. Utterance Length Distribution

Word Count Category	Scenarios	Accuracy (%)
Short (< 8 words)	5110	100.0%
Medium (8–18 words)	44186	100.0%
Long (> 18 words)	704	100.0%